

RefMetaPlant: a reference metabolome database for plants across five major phyla

Han Shi^{1,2,†}, Xueting Wu^{1,†}, Yan Zhu^{1,†}, Tao Jiang^{1,†}, Zhengwei Wang³, Xueting Li¹, Jianju Liu⁴, Yingxue Zhang⁵, Feng Chen⁶, Jinshan Gao⁷, Xiaoyan Xu⁸, Guoqing Zhang⁹, Ning Xiao⁴, Xianzhong Feng⁷, Peng Zhang^{10,2,10}, Yongrui Wu^{10,2,10}, Aihong Li⁴, Ping Chen^{1,*} and Xuan Li^{1,2,*}

¹Key Laboratory of Synthetic Biology, Key Laboratory of Plant Design, CAS Center for Excellence in Molecular Plant Sciences, Institute of Plant Physiology and Ecology, Chinese Academy of Sciences, Shanghai, China

²University of Chinese Academy of Sciences, Beijing, China

³Shanghai Chenshan Botanical Garden, Shanghai, China

⁴Institute of Agricultural Sciences for Lixiahe Region in Jiangsu, Yangzhou, China

⁵Shanghai Botanical Garden, Shanghai, China

⁶Agronomy College, Henan Agricultural University, Zhengzhou, China

⁷Key Laboratory of Soybean Molecular Design Breeding, Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun, China

⁸Core Facility Center, Center for Excellence in Molecular Plant Sciences, Institute of Plant Physiology and Ecology, Chinese Academy of Sciences, Shanghai, China

⁹National Genomics Data Center & Bio-Med Big Data Center, CAS Key Laboratory of Computational Biology, Shanghai Institute of Nutrition and Health, Chinese Academy of Sciences, Shanghai, China

¹⁰National Key Laboratory of Plant Molecular Genetics, Center for Excellence in Molecular Plant Sciences, Institute of Plant Physiology and Ecology, Chinese Academy of Sciences, Shanghai, China

*To whom correspondence should be addressed. Tel: +86 21 5492 4305; Email: lixuan@sippe.ac.cn

Correspondence may also be addressed to Ping Chen. Email: pchen@cemps.ac.cn

[†]The authors wish it to be known that, in their opinion, the first four authors should be regarded as Joint First Authors.

Abstract

Plants are unique with tremendous chemical diversity and metabolic complexity, which is highlighted by estimates that green plants collectively produce metabolites numbering in the millions. Plant metabolites play crucial roles in all aspects of plant biology, like growth, development, stress responses, etc. However, the lack of a reference metabolome for plants, and paucity of high-quality standard compound spectral libraries and related analytical tools, have hindered the discovery and functional study of phytochemicals in plants. Here, by leveraging an advanced LC–MS platform, we generated untargeted mass spectral data from >150 plant species collected across the five major phyla. Using a self-developed computation protocol, we constructed reference metabolome for 153 plant species. A ‘Reference Metabolome Database for Plants’ (RefMetaPlant) was built to encompass the reference metabolome, integrated standard compound mass spectral libraries for annotation, and related query and analytical tools like ‘LC–MS/MS Query’, ‘RefMetaBlast’ and ‘CompoundLibBlast’ for searches and profiling of plant metabolome and metabolite identification. Analogous to a reference genome in genomic research, RefMetaPlant provides a powerful platform to support plant genome-scale metabolite analysis to promote knowledge/data sharing and collaboration in the field of metabolomics. RefMetaPlant is freely available at <https://www.biosino.org/RefMetaDB/>.

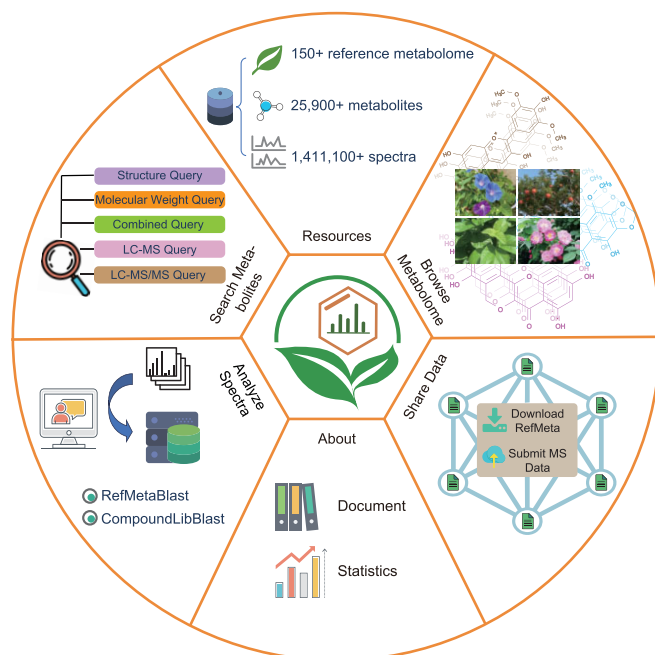
Received: August 13, 2023. Revised: October 13, 2023. Editorial Decision: October 13, 2023. Accepted: October 17, 2023

© The Author(s) 2023. Published by Oxford University Press on behalf of Nucleic Acids Research.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License

(<http://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com

Graphical abstract



Introduction

Metabolomics is defined as the science of qualitative and quantitative measurement of small molecule metabolites in living systems. The metabolome of a cell, a tissue, or an organism refers to the collection of all small molecule metabolites present in that biological system (1,2). Plants are unique in that they have developed their complex secondary metabolism through more than 900 million years of evolution. The astounding chemical diversity and metabolic complexity in plants are highlighted by estimates suggesting that metabolites in green plants number in the millions (3). These metabolites play crucial roles in various aspects of plant biology, including growth, development, and responses to environmental challenges (3,4). Plant metabolites contribute to the intricate web of biochemical reactions and signaling pathways that orchestrate plant physiology. In this context, the emerging field of plant metabolomics has garnered substantial attention, offering new opportunities to uncover the intricate metabolic networks in plants.

In recent decades, the study of plant metabolism has undergone significant advancements, largely driven by the emergence of metabolomics. Due to the enormous diversity and structural complexity of small molecule metabolites across plant species, their identification and functional analysis pose a significant challenge to researchers (5). The advance of metabolomics analytical technology, for example high-resolution mass spectrometry (HRMS) coupled with ultra-performance liquid chromatography (UPLC) separation, makes genome-scale metabolomic analysis possible, enabling detection of metabolites at an unparalleled sensitivity, resolution, accuracy, and throughput (3,4,6,7). Metabolomic technology based on high-performance of liquid chromatography combining high-resolution mass spectrometry is suitable for detecting the broad range of liquid-soluble metabolites found

in plants, thus has become a standard approach for plant metabolomics research.

Many databases and supporting tools related to metabolomics research and data sharing have been launched in the last two decades. Examples include MassBank (8), MassBase (9), HMDB (10), MetaboLights (11), Metabolomics Workbench (12), XMRs (13), RIKEN MetaDatabase (14), Metlin (15) and BinBase (16). While these resources provide many general tools for metabolomics data sharing and analysis, annotation with compound standard mass spectra, and profiling of MS-based metabolic data, they are not completely satisfactory when facing the many of the specific challenges of plant metabolomics research. One factor is that the rapid advance of MS technologies made many of the spectral repository data obsolete due to lower mass spectral qualities, i.e. accuracy, resolution, and sensitivity. Particularly, the shortcomings of the instruments and experimental conditions used to produce much of the existing MS data has resulted in limited coverage of metabolite peaks, reduced mass spectral quality, and lower analytical power. Furthermore, for plant metabolomics research, the lack of both reference metabolome and high-quality spectral data for standard compounds, as well as high-throughput analytical tools, has become a major bottleneck for the discovery and functional study of phytochemicals in plants.

To meet the needs of front-running phytochemical research, we have undertaken to address several challenges faced by plant metabolomics research: (i) We have generated untargeted mass spectral data, leveraging the most advanced LC-MS platform (UPLC-Q-Orbitrap-MS), from >150 plant species collected across five major phyla: Bryophyta, Lycopodiopsida, Pteridophyta, Gymnospermae and Angiospermae. (ii) We built a 'Reference Metabolome Database for Plants' (abbreviated as RefMetaPlant), using a computation protocol we developed for constructing a reference metabolome.

Inspired by the reference genome concept, the reference plant metabolome was constructed by collecting and ‘assembling’ its metabolites’ mass spectral data (MS1/MS2) from all available organs/tissues of the species. The first release of RefMetaPlant contains the reference metabolome for >150 plants, and has functions for metabolic research that were aimed to parallel that of RefSeq (17) for genomic studies. (iii) We have integrated standard compound mass spectral libraries for annotation, including both experimental reference mass spectra and *in silico* mass spectra, and developed a set of related query and analytical tools like ‘LC–MS/MS Query’, ‘RefMetaBlast’ and ‘CompoundLibBlast’ for plants metabolome searches and profiling, and metabolite identification.

It is worth noting that many existing metabolic databases predominantly focus on model organisms and a small number of plant species (18–21), which poses significant limitations that hinder the exploration of the metabolomics space in the broad field of plant research. Therefore, we have established RefMetaPlant as a broader more integrated platform to enable increased interactions between scientists from different parts of the world and ease the flow of information within the scientific community (22). RefMetaPlant serves as an integrated database and analysis platform dedicated to becoming the centralized resource for plant metabolomic research. It aims to standardize and integrate reference metabolome data, providing a comprehensive platform for researchers in plant metabolomics, genetics, and related fields. RefMetaPlant is freely available at <https://www.biosino.org/RefMetaDB/>, which offers a user-friendly web interface and a variety of easy-to-use tools for the functions mentioned above. This comprehensive platform caters to a broad audience, including scientists in plant metabolomics research, geneticists in plant breeding, and even consumers interested in understanding the chemical contents of plants. RefMetaPlant was established to accelerate advancements in plant metabolomics research, foster collaborations and promote a deeper understanding of plant metabolism.

Materials and methods

Plant materials

To broadly cover terrestrial plants, we collected plant samples from various sources (Supplementary Table S1). A total of 153 plant species have been collected so far, with each species comprising one or more tissues, like leaf, flower, seed, stem or root. Three biological replicated samples were obtained for each tissue. The collected samples were immediately frozen in liquid nitrogen before stored at -80°C . For study cases 1 and 2, rice and maize samples were harvested in Jiangsu (Yangzhou, southeastern China) and in Hainan (Sanya, southern China), respectively.

DNA extraction, PCR amplification, ITS sequencing and annotation

Fresh leaf tissue was first ground into powder with liquid nitrogen. Total genomic DNA was subsequently extracted using the Plant Genomic DNA Kit (Tiangen Biotech Co., China) according to the manufacturer’s instructions. The plant internal transcribed spacer 2 (ITS2) region was amplified using the following primers: ITS2-2F (5′-ATGCGATACTTGGTGTGAAT-3′) and ITS2-3R (5′-GACGCTTCTCCAGACTACAAT-3′). PCR was performed in 50 μl reaction mixtures, containing

5 μl of 10 $\times$ Buffer (including 2.5 mM Mg^{2+}), 1 μl of Taq polymerase (TaKaRa), 1.5 μl of each PCR primer (10 μM), 1 μl of dNTP (10mM) and 1 μl of DNA extract (20ng/ μl), and the total volume was adjusted to 50 μl with sterile deionized water. PCR amplification was conducted according to the following procedure: 95°C for 5 min, 35 cycles of 95°C for 30 s, 58°C for 30 s and 72°C for 1.5 min, and a final extension at 72°C for 7 min. PCR products were analyzed by electrophoresis in a 1% agarose gel. The PCR products were purified using an AxyPrep DNA Gel Extraction Kit (AXYGEN, Hangzhou, China) and sequenced bidirectionally using a ABI 3730XL sequencer (Applied Biosystems Co., USA) based on the Sanger sequencing method at Shanghai Personalbio Technology Co., Ltd. The sequenced ITS2 sequences were aligned against the NCBI NT database with the megablast tool (23), and the species with the highest sequence similarity was selected.

Metabolite extraction and mass spectral analysis

The samples were processed for mass spectral analysis as described in our previous study (24). In brief, the freeze-dried tissues were ground using a tissue grinder (No. 05010997; Shanghai BiHeng Biotechnology Company Limited, Shanghai, China) for 2 min at 55 Hz. 150 mg of powder was weighed and mixed with 1.5 ml 70% aqueous methanol (containing 1 mg/l capsaicin as the internal standard). The samples were vortexed every 10 min three times and extracted overnight at 4°C . Following centrifugation at 12 000 g for 15 min, all the supernatants were filtered using a 0.22 μm filter and transferred to a liquid phase bottle for subsequent analysis.

Samples were analyzed on a Waters Acquity Ultra Performance Liquid Chromatography system equipped with an ACQUITY UPLC BEH C18 column (1.7 μm , 2.1 mm $\times$ 100 mm) coupled with Q Exactive hybrid Q-Orbitrap-high resolution mass spectrometry (Q-Orbitrap-HRMS) (Thermo Fisher Scientific). The instrument parameter settings used for the analysis followed the procedure outlined in our previous study (24).

Workflow for assembling the reference metabolome

The workflow for constructing the plant reference metabolome consisted of four major steps (Figure 1B): (i) Raw MS data processing: Raw MS files were processed in Compound Discoverer software (version 3.2, Thermo Scientific) with automatic workflow for LC spectra alignment by retention time (RT) and peak detection. To refine the metabolite features, a set of criteria were applied (25), including reproducibility <90%, sample to blank ratio <10%, peak area less than $1\text{E}5$, and elimination of features exhibiting multi-ion adducts (such as Na^+ , K^+ , NH_4^+ , Cl^-), isotopes, in-source fragmentation, or dimerization. (ii) Assembly of reference metabolome: an in-house script based on Xcalibur Development Kit (XDK) in Xcalibur software was first used to automatically acquire MS1/MS2 paired data from the raw LC–MS/MS files derived from distinct tissues of the plant species. The repetitive features identified from these diverse tissues were merged to establish the reference metabolome, encompassing spectral features separately in both ESI positive and negative ion mode. The merging process adhered to the following parameters: mass tolerance 5 ppm; retention time tolerance 0.2 min. (iii) Annotation of reference metabolome: the annotation of the reference metabolome was carried

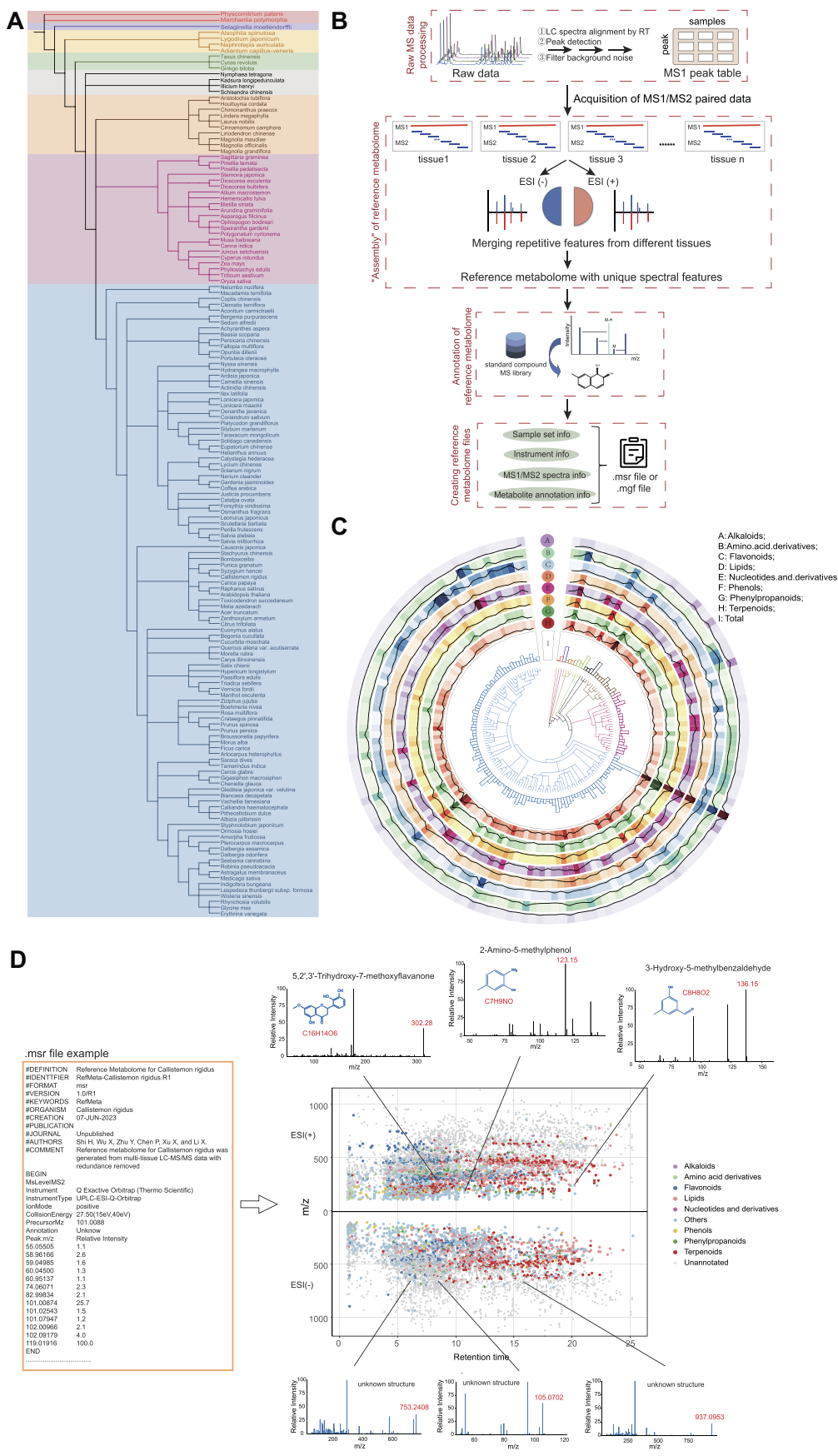


Figure 1. Integration of plant species and metabolome data. (A) A taxonomic tree depicting the taxonomic relationships among 153 plant species that are representative of the five major phyla: Bryophyta (red), Lycopodiopsida (purple), Pteridophyte (yellow), Gymnospermae (green) and Angiospermae.

out following our prior research (24). In summary, we employed two complementary approaches for this process. Initially, we searched against the experimental reference mass spectra library obtained from public databases and structurally annotated MS2 spectra. Subsequently, we utilized predicted theoretical spectra to further enhance the scope of the structural annotations. To achieve a hierarchical chemical classification of the compounds, we utilized the Ruby application programming interface within the ClassyFire software (26) to generate the chemical classes. (iv) Creating reference metabolome files: sample, instrument, MS data and annotation information were integrated to generate reference metabolome file in .msr or .mgf format.

Construction of standard compounds mass spectral libraries

The standard compounds mass spectral libraries consist of experimental MS/MS spectra and *in silico* spectra. The experimental MS/MS spectra data were sourced from a range of databases including MassBank (8), MoNA (<https://mona.fiehnlab.ucdavis.edu/>), ReSpec (27), FiehnLib (28) and RIKEN PlaSMA (<http://plasma.riken.jp/>). According to the use of specific keywords and meticulous filter criteria, the mass spectral data corresponding to the metabolites of interest under investigation were retrieved. For each individual metabolite, we procured its corresponding mass spectrum, molecular formula, molecular weight and potential metabolite information from these databases.

The *in silico* mass spectral data were generated by converting the structural information of biologically relevant compounds into theoretical spectrograms using CFM-ID software (29). The structural data for each compound were collected from four biologically relevant structure databases: KEGG (30), KNApSACk (31), PubChem (32) and UNPD (33). For the KEGG database, '2Dmol' files obtained via the 'python-Bio.KEGG' API were utilized to fill in missing structural information through the application of OpenBabel software (34), as SMILES, InChIKey and InChI were not available. In the case of the PubChem database, compounds from the PubChem classification browser, annotated with NCBI BioSystems (35), biological role classification from ChEBI Ontology (36), and classifications as flavonoids or prenol lipids from LIPID MAPS structure database (37) were downloaded.

Implementation of structure query and spectral query

The structure query was implemented via the function 'Search metabolites' using the 'Structure Query' tool. The JavaScript based molecule editor JSME (38) was employed to transform the drawn molecule structure into a canonical SMILES format. Subsequently, the SMILES representation of the query molecule was converted into an InChI string using the chem-

istry development kit (39). This InChI string was then utilized for a query against the plant reference metabolome to identify matching hits.

For the spectra query, the function 'Search metabolites', particularly in the 'LC-MS/MS Query', as well as related functions such as RefMetaBlast and CompoundLibBlast analytical tools, the INCOS algorithm (40) was employed through java scripts to score the similarity between reference mass spectra. In brief, the reference mass spectra that have similar precursor m/z (within a user-defined precursor mass tolerance range) with query mass spectra were retrieved and compared for similarity using the INCOS algorithm. Reference mass spectra with a similarity score >0.75 were used to assign annotation to query mass spectra.

Implementation of analytical tools 'RefMetaBlast' and 'CompoundLibBlast'

The RefMetaBlast and CompoundLibBlast analytical tools, which can be used via the function 'Analyze Spectra', are designed for LC-MS/MS data annotation. These tools are intended for use with high-resolution LC-MS/MS data acquired in centroid mode and provided in standard formats such as mzML, mzXML or mzData. Upon submission of an annotation task, the LC-MS data undergoes initial processing using a custom R script built upon the XCMS package (41). This processing encompasses peak detection, peak filtering, and extraction of MS2 spectra. Subsequently, each MS2 spectrum is annotated by a database search via the INCOS algorithm. The RefMetaBlast tool can be used to analyze a sample's spectra with a reference metabolome from a specific species. The CompoundLibBlast tool analyzes a sample's spectra with public mass spectral data and *in silico* mass spectral data. In addition, the result of matched or annotated metabolites can be conveniently downloaded in both JSON and CSV formats.

Database and web interface implementation

The RefMetaPlant project was developed and integrated on the Java SSM framework (Spring, Spring MVC and MyBatis) and the packaged application was deployed on an Apache Tomcat web server (<http://tomcat.apache.org/>) which runs on the CentOS 6.8 system. The interactive RefMetaPlant web interface was developed using Bootstrap (3.3.7, <http://getbootstrap.com/>) and jQuery (2.1.0, <http://jquery.com/>) and implemented in JSP. All the data in RefMetaPlant were stored in a relational database MySQL (5.1.73, <http://www.mysql.com/>). The 2D and 3D structure of compounds were displayed using Smiles Drawer and JSmol (<http://wiki.jmol.org/index.php/JSmol>), respectively. The JavaScript plotting library Flot (<http://www.flotcharts.org/>) was used to draw interactive mass spectrum. The interactive result tables for annotated metabolites were generated using DataTables (<https://www.datatables.net/>). The back-end logic processing and data anal-

The color scheme for subgroups of Angiospermae: ANA grade, grey; Magnoliids, orange; Monocotyledons, magenta; and Dicotyledons, blue. (B) Workflow for constructing plant reference metabolome, which consists of four major steps: (i) raw MS data processing; (ii) assembly of reference metabolome; (iii) annotation of reference metabolome and (iv) creating reference metabolome files (detailed description in Materials and Methods). (C) Global metabolic profiles of the reference metabolome for 153 plants. The innermost circuit depicts the taxonomic relationships of the 153 species. Circuits A–H display the profiles of metabolites belonging to eight chemical classes (A = alkaloids, B = amino acid derivatives, C = flavonoids, D = lipids, E = nucleotides and derivatives, F = phenols, G = phenylpropanoids, H = terpenoids). The intensity of color shading in circuits A–H indicates the number of metabolites detected for each species. The bar chart in circuit I represents the total number of detected metabolites. (D) The reference metabolome for *Callistemon rigidus*. The header section of .msr file is illustrated to the left. Metabolites are represented as dots, with chemical classifications indicated by their color. Examples of annotated and unknown metabolites are presented at the top and bottom, respectively.

ysis were mainly implemented through R and Java scripts. The database is available online without registration and is optimized for Chrome, Opera, Firefox, Windows Edge and macOS Safari.

Statistical analysis

Statistical analyses were mainly conducted using R software (<https://www.R-project.org/>). Affinity Propagation (AP) clustering was employed to group metabolites with similar abundance profiles utilizing the 'apcluster' package in R (42). The taxonomic relationships among species were generated from the NCBI Taxonomy database (43), and visualized using the interactive Tree of Life (iTOL) web server (44). Other data analyses were carried out by in-house scripts written in R or Perl. The manuscript figures were mostly drawn by R.

Results

Constructing a reference metabolome for plants

A reference metabolome refers to the global collection of all metabolites for a given species. An analog comparison between the reference metabolome and metabolites for a given species mirrors that between the genome and genes for the species. In this study, the reference metabolome for a plant species is constructed by collecting and 'assembling' its metabolites' mass spectral data (MS1/MS2) from all available organs/tissues of the species. Nowadays the high-resolution mass spectrometry can detect and identify a massive number of heterogeneous phytochemicals that have highly diverse structures and chemical and physical properties, usually numbering in the tens of thousands. Thus, in order to construct a plant reference metabolome an efficient computing process must be implemented to handle and resolve the mass spectral data, to eliminate redundant features, and to annotate detected metabolites.

To broadly cover terrestrial plants, we have so far collected tissue samples from more than 150 plant species that are representative of the five major phyla: Bryophyta, Lycopodiopsida, Pteridophyta, Gymnospermae, and Angiospermae (Figure 1A, and Supplementary Table S1). Their classifications were validated by ITS sequencing on each plant sample (Supplementary Table S2).

Mass spectral analysis was performed using ultra-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) platform on the tissues collected for each plant species. The raw spectral data were processed and filtered to obtain high-quality feature signals (Figure 1B). To assemble the reference metabolome for each plant species, MS1/MS2 spectra acquired from different tissue samples were merged for ESI positive and negative ion modes, by removing duplicated features. Then unique spectral features were annotated using compound spectral libraries before reference metabolome files (in .msr or .mgf format) were created for each species (Figure 1B).

The first release of the reference metabolome illustrates the metabolic diversity and complexity of each of the 153 plant species (Figure 1C, and Supplementary Table S3). Notably, annotated metabolites are classified into eight groups, i.e. alkaloids, amino acid derivatives, flavonoids, lipids, nucleotides and derivatives, phenols, phenylpropanoids, and terpenoids. As an example, the reference metabolome of *Callistemon rigidus* is depicted in Figure 1D. It contains MS1/MS2

data for all metabolites detected in *C. rigidus*, in both positive and negative ion modes, and, if available, annotation of the spectral features. Metadata can be found in the header section of the reference metabolome file (.msr or .mgf format). Like its reference genome, the reference metabolome of *C. rigidus* provides an essential reference for studies of its metabolism and physiology.

RefMetaPlant: a reference metabolome database and related toolset for plant metabolic research

Based on the reference metabolome constructed for plants of the five major phyla, we have developed and released the first version of a reference metabolome database for plants, RefMetaPlant. We use RefMeta for its abbreviation, to parallel with 'RefSeq', the reference sequence database at NCBI (23). RefMeta contains six major components, including data resources, database documentation and statistics, and 'Browse Metabolome', 'Search metabolites', 'Analyze spectra' and 'Share data' tools (Figure 2A, Supplementary File S1).

RefMetaPlant utilizes a large collection of novel and existing data, and data resources that are growing daily: (i) >1 086 000 experimental mass spectra obtained in this study using UPLC coupled with Quadrupole-Orbitrap High Resolution Mass Spectrometer (UPLC-Q-Orbitrap-HRMS) on samples of >150 plant species from Bryophyta, Lycopodiopsida, Pteridophyta, Gymnospermae and Angiospermae (Figure 2B, top panel); (ii) a library of >383 700 biologically relevant compounds, which includes their structural and property information that was collected from public databases like KEGG (30), KNApSack (31), PubChem (32) and UNPD (33) (Figure 2B middle panel); (iii) a library of >325 100 mass spectral data for standard compounds which includes 135 464 experimental reference mass spectral from public databases like MassBank (8), MoNA (<https://mona.fiehnlab.ucdavis.edu/>), ReSpec (27), FiehnLib (28), RIKEN PlaSMA (<http://plasma.riken.jp/>) and data of 189 639 *in silico* mass spectra (29) (Figure 2B, bottom panel); (iv) the reference metabolome (in .msr and .mgf formats) that we constructed for each of the 153 plant species (Supplementary Table S4). These data are available for download and for analysis applications using our online tools.

Analytical functions implemented in RefMetaPlant

The function 'Browse Metabolome' allows users to browse the reference metabolome of selected plant species and further drill into the details of each metabolite, including its formula, classification, 2D/3D structure, exact mass, SMILES, spectra and related metadata, etc. (Figure 3A). The sample and analytical method information of each plant species can also be found from 'metadata' link in the 'species metabolite' page. In addition, the reference metabolome of selected plant species can be downloaded in files (two different formats: .msr or .mgf format) via the function 'Share data'.

The function 'Search metabolites' supports querying the reference metabolome using five different types of queries, i.e. structure sketch, molecular weight, combined parameters, LC-MS and LC-MS/MS (Figure 3B). For the latter two, queries can be submitted with a list of precursor ion (MS1), or a parent ion (MS1) with its fragment ions (MS2), and with other options for specified plants species, tolerance of *m/z*, or library for query. As a result, a list of matched metabolites is returned, which allows users to view the details of each

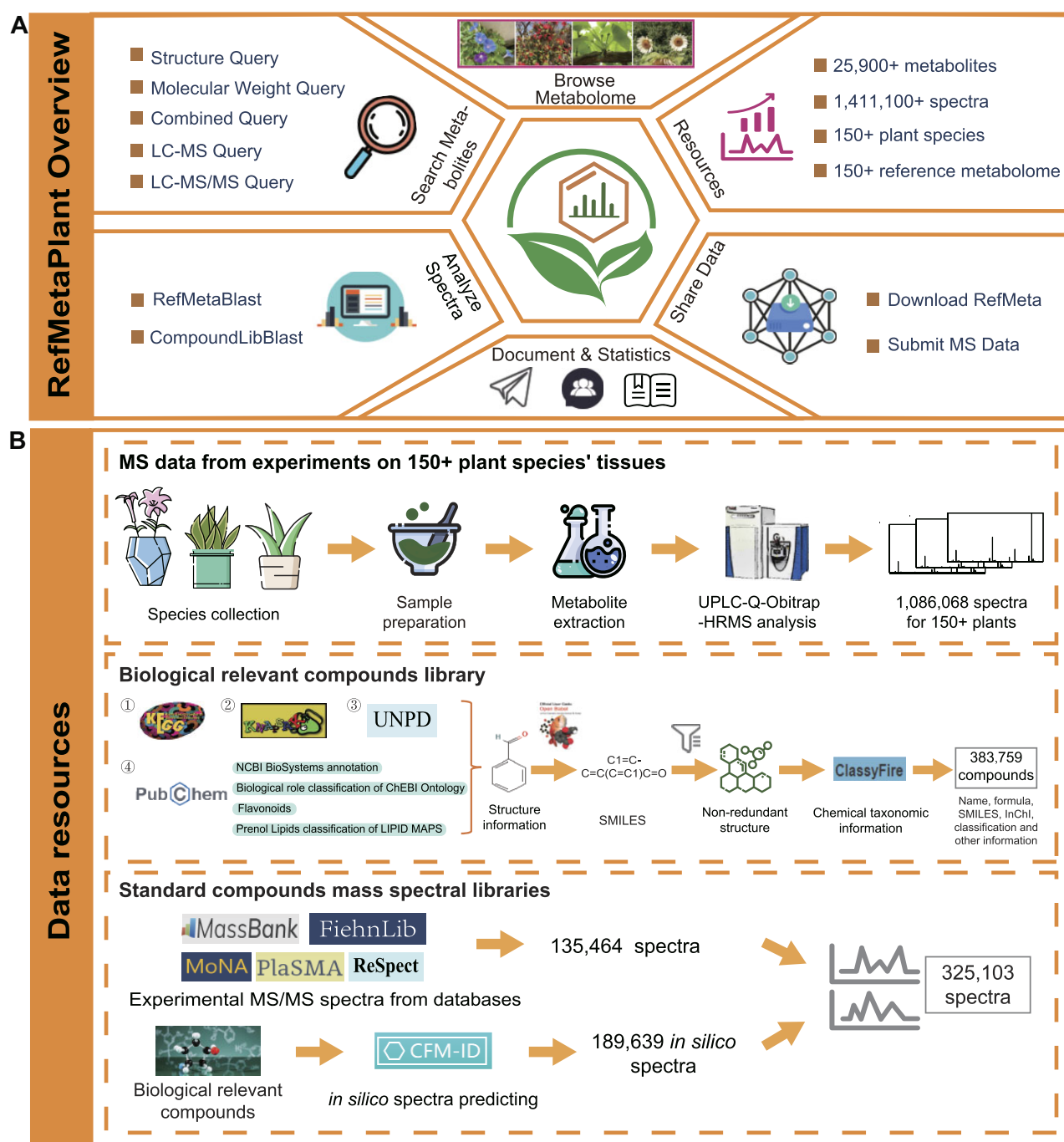


Figure 2. Overview of RefMetaPlant and its resources. **(A)** Overview of RefMetaPlant. RefMetaPlant comprises six major components: 'Browse Metabolome', 'Search Metabolites', 'Analyze Spectra', 'Resources', 'Share Data' and 'Document & Statistics'. **(B)** Data resources and their related workflow for RefMetaPlant. The resource data of RefMetaPlant encompass the MS data ($n = 1\,086\,068$) from experiments on 150+ plant species' tissues (top); a biological relevant compounds library ($n = 383\,759$) (middle) and standard compounds mass spectral libraries ($n = 325\,103$) (bottom).

metabolite by following a link. Importantly, all plant species that contain the metabolite in their reference metabolome are shown, revealing the 'popularity' of the metabolite. In addition, a filter box is also provided for the above tools for users to select data from the list of returned results by typing in a parameter, such as name, molecular weight, SMILE, classification, etc.

To facilitate the use of RefMeta database for plant metabolic research, we developed two related analytical tools, RefMetaBlast and CompoundLibBlast (Figure 3C).

RefMetaBlast analyzes a sample's spectra by comparing it with a selected reference metabolome for species-specific metabolic profiling. The reference metabolome selected for comparison provides a high-confidence reference for profiling of interested samples by users. The function 'CompoundLibBlast' analyzes a sample's spectra with selected compound libraries for automated metabolite annotation and identification. Its function is performed based on the experimental reference mass spectral data from public databases, and the *in silico* mass spectral data we generated (24,29). These spectral

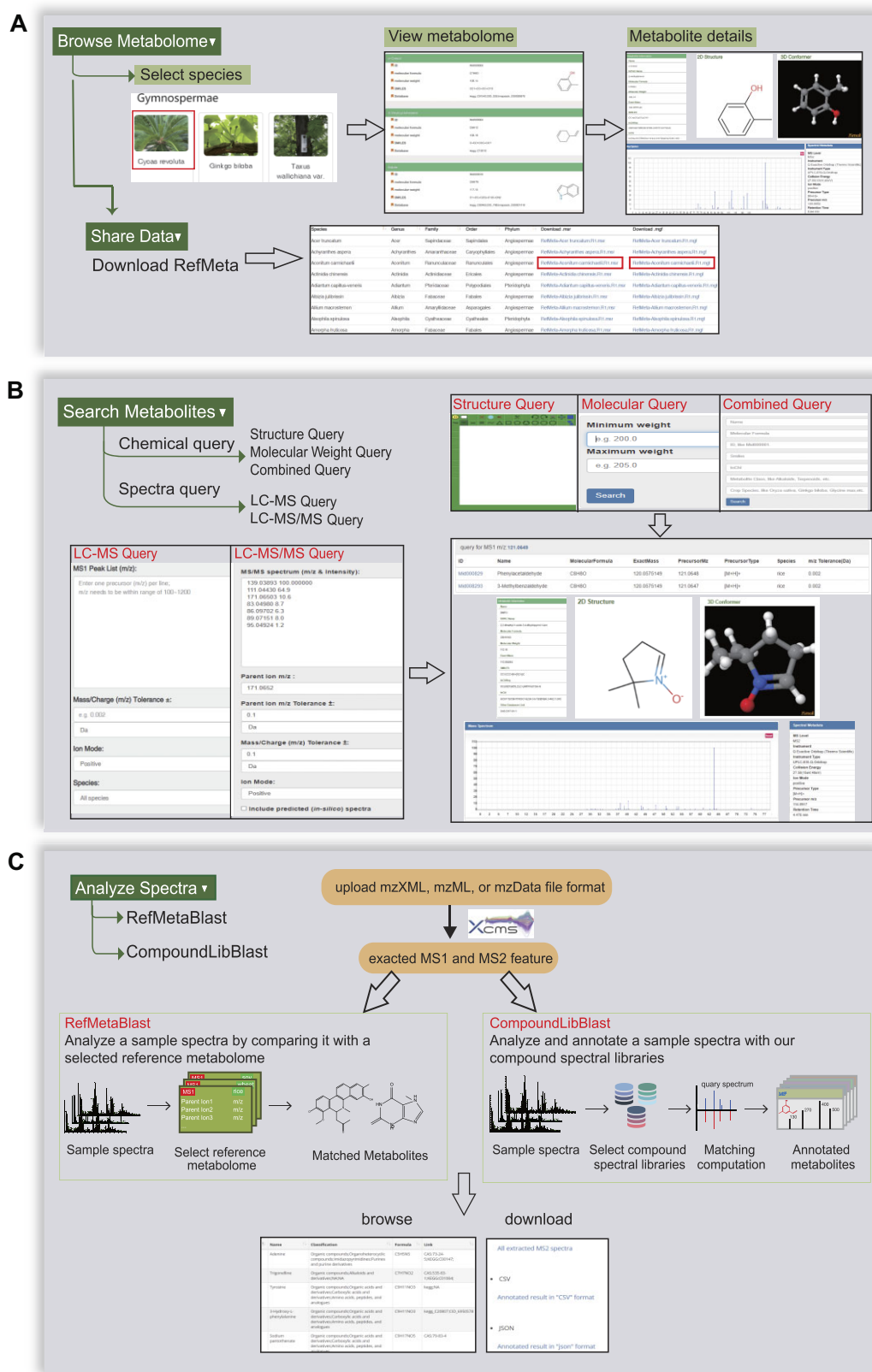


Figure 3. Implementation of analytical functions in RefMetaPlant. **(A)** Overview of function 'Browse Metabolome'. The metabolome of each species contains metabolites with structural and spectral information. The reference metabolome of selected species can be downloaded in files via the function 'Share data' in .msr or .mgf format. **(B)** Overview of function 'Search Metabolites' enabling multiple ways to query reference metabolome. Function for querying metabolites across chemical query supporting both structure sketch, molecular weight, combined parameters. RefMetaPlant also supports the identification of unknown mass spectral data by querying LC-MS and LC-MS/MS. The query results showed the structure information and spectral information of each metabolite. **(C)** Overview of functions 'Analyze Spectra' supporting analytical tools 'RefMetaBlast' and 'CompoundLibBlast'. Their workflows are described in details in Materials and methods.

analytical tools are implemented through an interactive webpage with some advanced features that were not available before in an open-source tool. Query data in mzXML, mzML or mzData format are uploaded first. Data analysis is initiated by clicking the 'Start analysis' button, and a hyperlink is provided for long running analysis, to allow users to retrieve the results. Both RefMetaBlast and CompoundLibBlast functions return results of matched or annotated metabolites in addition to downloadable files containing extracted MS2 spectra (MGF format), and annotation data (CSV and JSON format) (Figure 3C).

Case study 1: metabolic trends during the rice grain filling period

Grain filling is the growth stage in rice when fertilized ovaries acquire carbohydrate and other nutrients to synthesize starch and convert themselves into grains (45,46). It determines the yield and quality of cereal crops (47). Accumulation of metabolites in filling grains involves the coordination of different metabolic and cellular processes, understanding the dynamic changes of these substances, and is of great significance in study of seed development and crop breeding (48).

We applied RefMetaBlast to analyze the metabolic profile of *Oryza sativa* var. Yangdao #6 during the process of grain filling. Rice grains from 10, 20 and 30 days after flowering (DAF), were harvested and processed for mass spectrum experiments (Figure 4A). Raw mass spectral data were extracted with ProteoWizard (49) before they were assigned using the rice reference metabolome and the online tool RefMetaBlast. Consequently, 468, 469 and 511 metabolites were detected above the threshold for the 10, 20 and 30 DAF grain samples, respectively (Supplementary Table S5). The numbers of different metabolites identified in the rice grains varied between different developmental stages (Figure 4B). Lipid compounds were found to be the most abundant, followed by terpenoids, and flavonoids. Their numbers were generally trended higher at later growth stages except for amino acid derivatives, alkaloids and flavonoids.

For the grain metabolites assigned by RefMetaBlast, their relative abundance was grouped with affinity propagation (AP) clustering to investigate their changes in their patterns. The changes of metabolite abundance were grouped into nine distinct clusters (Figure 4C, D). Notably, about 46% of amino acid derivatives belonged to cluster II, which showed a trend of gradual reduction between 10 DAF and 30 DAF. A substantial number of terpenoids first increased between 10 DAF and 20 DAF, then decreased between 20 DAF and 30 DAF (cluster III, cluster VI and cluster VII; Figure 4C, D). Lipids appeared to exhibit temporal specificity, with different types of lipids present at different time points. We observed a significant proportion of lipids in Clusters II (~25%), which was high at 10 DAF, III (~17%), which was high at 20 DAF, and VIII (~21%), which was high at 30 DAF. About 15% of the flavonoids were found to increase from 10 DAF to 30 DAF (clusters VIII and XI). Notably, rhoifolin and apigenin 4'-glucuronide, two derivatives of apigenin that belong to flavonoid family, were found to only be present at 30 DAF. As they share a chemical structure resembling that of anthocyanins, it remains to be determined if they play a role in the yellowing process in rice grains. The similar kinetics of different groups of metabolites during rice grain development may suggest common regulation mechanisms for the related pathways, which can help direct further study. The above findings demonstrate the utility

of RefMetaBlast as a tool, enabling a deeper understanding of the changes and regulatory mechanisms of metabolites during the growth and development of rice.

Case study 2: discovery and expanded annotation of metabolites in maize samples

The CompoundLibBlast tool was designed to annotate plant metabolites using expanded compound mass spectral libraries, including both experimental and *in silico* mass spectral data (24,29). To demonstrate its performance, we conducted an analysis of non-targeted metabolomic data for corn samples of *Zea mays* var. B73 (Figure 4E), using CompoundLibBlast. We identified a total of 746 non-redundant metabolites, which belonged to eight chemical classes (Figure 4F, Supplementary Table S6). Among these, the largest class of metabolites was lipids, which contained 170 compounds, highlighting the richness of lipids in corn.

To evaluate the performance of CompoundLibBlast, we compared its results with those reported in recently published studies (Figure 4G). The number of metabolites annotated by our tool increased significantly compared to those of the early studies (50–54). We further compared our results with that of Liang and colleagues' (54), which reported the most metabolites among previous studies. Our tool, CompoundLibBlast, was able to identify additional metabolites in corn, particularly some nutrition-related substances, such as vitamin K1, riboflavin, and linoleic acid, which were not found in the 2021 study (54) (Figure 4H). Thus, CompoundLibBlast is shown to capture a significantly broader range of metabolites, identifying many that were previously overlooked. However, further experimental validation is necessary to establish the identities of these metabolites and their functions in biological contexts. This case proved CompoundLibBlast to be a valuable tool to expand the study of diverse metabolites in crops, offering great potentials for discovery of novel metabolites and for a deeper understanding of metabolism in plants.

Case study 3: specialization of secondary metabolites in the evolution of legumes

The evolving functions of secondary metabolites in plants has been subjected to intense study (55–57). Legumes belong to the family Leguminosae, which is the third-largest plant family. It has traditionally been divided into three notable subfamilies, Caesalpinioideae, Mimosoideae, and Faboideae (58). We have collected samples from 27 legume plants across all three subfamilies, for which a reference metabolome was constructed (Figure 5A, left panel). Their global metabolic profiles revealed variable number of metabolites for each chemical class between these legume species (Figure 5A, right panel, Supplementary Table S7). Notably, while lipids and flavonoids are the largest chemical categories in legumes, the numbers of lipids and flavonoids fluctuated the most in closely related species, despite their shared family affiliation.

We then took a closer look at flavonoid biosynthesis in the legume subfamilies. In the flavonoid biosynthesis pathway, the C6–C3–C6 carbon skeleton is formed (in 4-coumaroyl-CoA) from phenylalanine (Figure 5B). Then the basic flavonoid skeleton is modified by different groups of enzymes, leading to different biosynthesis pathways, and the production of many important intermediates (59–61). Isoflavonoids can

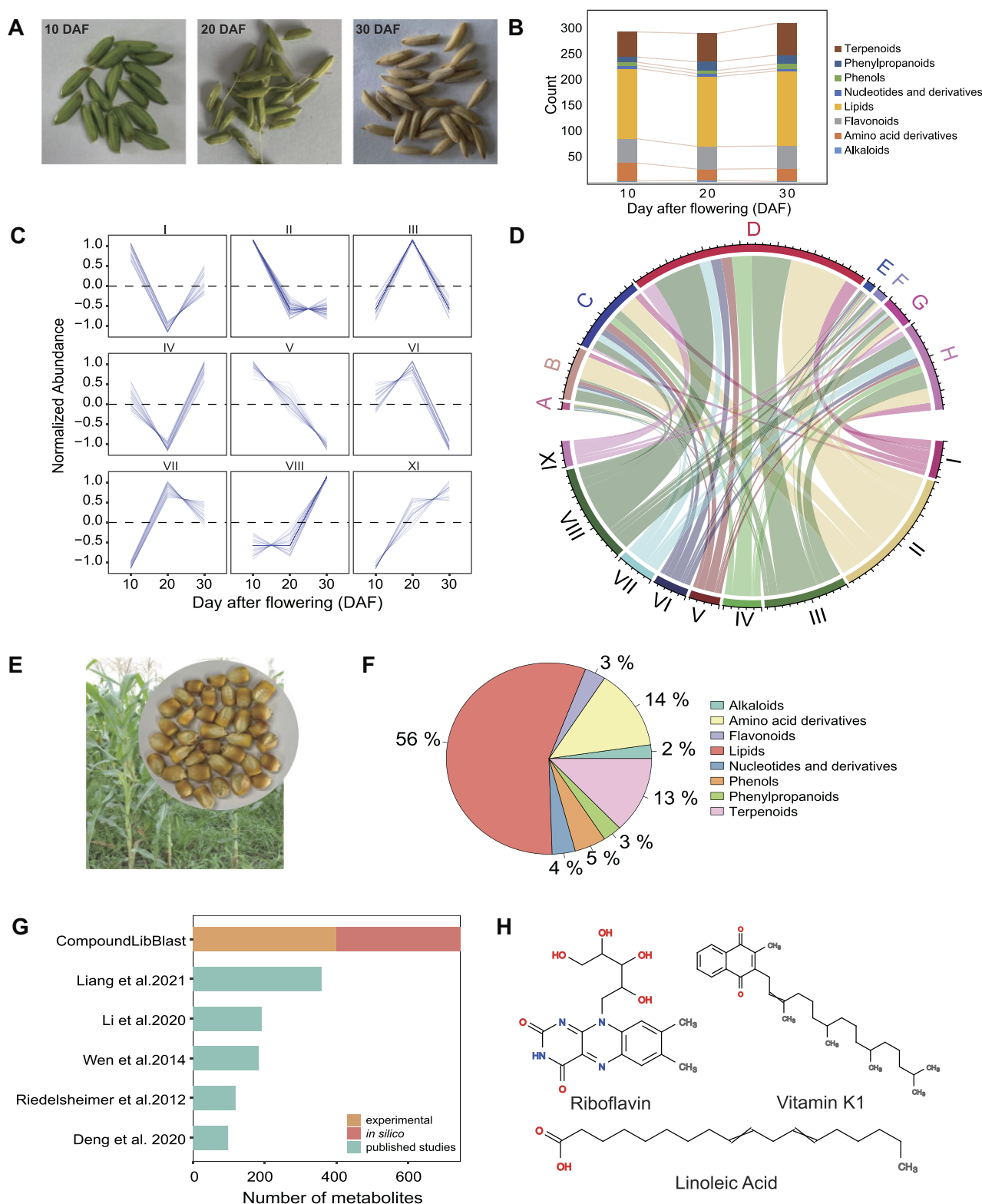


Figure 4. Case study of rice and corn grains utilizing RefMetaBlast and CompoundLibBlast. **(A)** Grains of *Oryza sativa* var. Yangdao #6 obtained at 10, 20 and 30 days after flowering (DAF). **(B)** The numbers of metabolites identified with RefMetaBlast at various stages of grain filling. Chemical class for metabolites is color-coded as indicated to the right. **(C)** Affinity propagation (AP) clustering of metabolite groups in similar temporal changes along grain filling process of *Oryza sativa* var. Yangdao #6. Each cluster represents metabolites with similar abundance profiles. **(D)** Circos diagram illustrating the chemical class annotation for metabolite clusters I-IX that correspond to those in (C). Chemical class labeling: A = Alkaloids, B = Amino acid derivatives, C = Flavonoids, D = Lipids, E = Nucleotides and derivatives, F = Phenols, G = Phenylpropanoids, H = Terpenoids. **(E)** Grains of *Zea mays* var. B73. Corn grains were harvested in March 2021 from Sanya, Hainan, and subsequently used for mass spectral analysis. **(F)** Metabolite compositions for grains of *Zea mays* var. B73 identified using CompoundLibBlast. **(G)** Comparison of metabolites identified in B73 corn grains between our results (CompoundLibBlast) and previous five studies (50–54). **(H)** Example of compounds in *Zea mays* var. B73 newly identified by CompoundLibBlast but not in previous study (54).

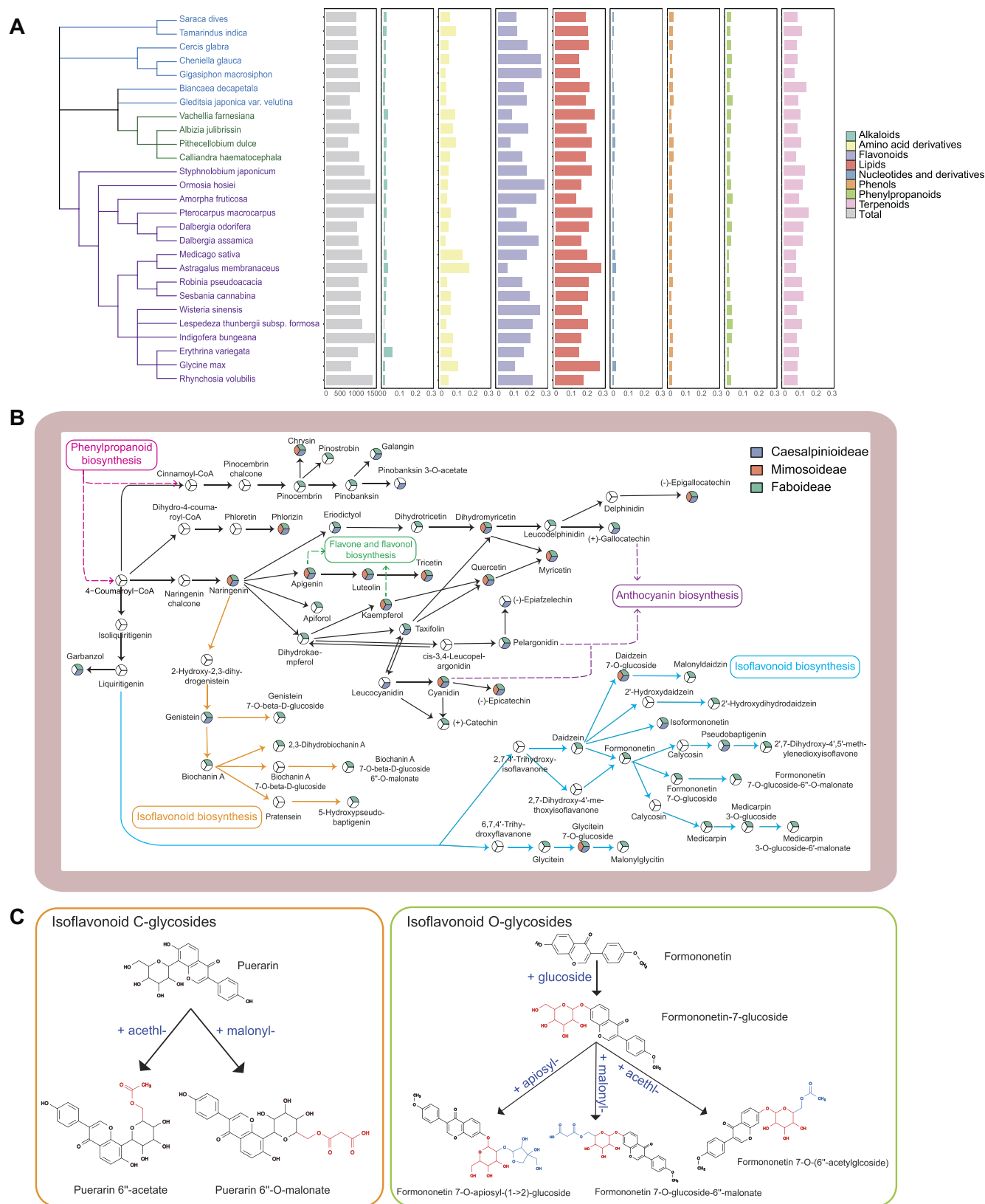


Figure 5. Reference Metabolome revealing specialization of secondary metabolites in evolution of legumes. **(A)** Taxonomic relationships and the global metabolic profiles of 27 legume species. Legume subfamilies are color-coded, namely Caesalpinioideae (blue), Mimosoideae (green) and Faboideae (purple). The distribution of metabolite class for each species is depicted to the right. **(B)** Distribution of metabolites in the flavonoid biosynthesis pathway in three legume subfamilies. The presence of metabolites within the pathways is indicated by colored shaded fan areas. Metabolites are labeled with their respective KEGG designations. **(C)** C- and O-glycosylated isoflavonoids in Faboideae. Derivatives of C-glycosylated isoflavonoids can be modified by acetyl- or malonyl- groups (left panel), and derivatives of O-glycosylated isoflavonoids can be modified by acetyl-, malonyl- or apiosyl- groups (right panel).

be synthesized through two different pathway branches (Figure 5B, the yellow and blue branch). Strikingly, we found major differences in isoflavonoid biosynthesis between the three subfamilies, Caesalpinioideae, Mimosoideae, and Faboideae. Based on all the species we collected and examined, the isoflavonoid biosynthesis pathways (Figure 5B, the yellow and blue branch) were found to exist uniquely in species belonging to Faboideae. Our findings add further evidence to the restricted occurrence of isoflavonoids in the subfamily Faboideae, which is suggested to be a remarkable feature of the chemistry of the Leguminosae (62). The phenomena of specialized metabolites restricted to subfamilies, tribes and genera, is emphasized in a term called chemotaxonomy, for which chemical profiles were used as a systemic framework to reflect phylogeny (63). Our results revealed that metabolic profiles varied across the sampled legume species, with isoflavones having a restricted distribution compared to flavones and anthocyanins, which were more broadly distributed. Isoflavonoids serve numerous important physiological functions, including as anti-oxidants and phytoalexins (64,65). The data suggest that flavonoids, particularly isoflavonoids, have been strongly influenced by selection and adaptation during plant evolution.

We further examined ~400 isoflavonoids that are specifically present in Faboideae. These metabolites were C-glycosylated and O-glycosylated isoflavonoids, which can be further modified by acetyl-, malonyl- and apiosyl- groups. For example, the C-glycosylated isoflavonoid puerarin can be modified by acetyl-, and malonyl- groups, leading to production of puerarin 6''-acetate and puerarin 6''-O-malonate, respectively (Figure 5C, left panel). On the other hand, formononetin can be modified by a glucoside, and subsequently by acetyl-, malonyl- or apiosyl- groups to form different isoflavone derivatives (Figure 5C, right panel). Our findings added new members to the list of O-glycosylated isoflavonoids with malonyl-modifications reported for legume species (63). For plant natural products, glycosylation is a versatile modification used to enhance their solubility and stability, and promote storage in plant cells (66). It also affects bioactivity and bioavailability of natural products (67). Taken together, these findings provide valuable insights into the ecological adaptation and evolution of legumes.

Summary and future directions

In recent decades, study in the field of plant metabolism has advanced rapidly, largely driven by metabolomics research, and has revealed an astounding amount of chemical diversity in this ancient kingdom. For a given species, the metabolome and metabolites of a species is analogous the genome and genes, and in the latter case, a reference genome provides a valuable foundational resource. However, the lack of a reference metabolome for plants, and deficits of high-quality standard compound spectral libraries and high-throughput analytical tools, have hindered the discovery and functional study of phytochemicals. The ideal reference metabolome for a plant species consists of mass spectral data (MS1/MS2) for all its metabolites across all organs/tissues of the species. With this goal in mind, we have undertaken efforts to address the specific challenges in plant metabolomics research. First, by leveraging the most advanced LC-MS platform, we generated untargeted mass spectral data over a period of five years, from >150 plant species across the five major phyla

of green plants. Second, we built RefMetaPlant, a 'Reference Metabolome Database for Plants'. Using a computation protocol that we developed for constructing reference metabolomes, the first release of RefMetaPlant contains reference metabolomes for 153 plant species. Third, RefMetaPlant has designated functions for metabolic research that were aimed to parallel that of RefSeq for genomic studies. In the RefMetaPlant platform, we have integrated standard compound mass spectral libraries to assist annotation, including both experimental reference mass spectra and *in silico* mass spectra. We have also developed a set of related query and analytical tools like 'LC-MS/MS Query', 'RefMetaBlast' and 'CompoundLibBlast' for plant metabolome searches and profiling, and metabolite identification. Thus, RefMetaPlant provides a powerful platform to support plant metabolome research and genome-scale metabolomics analysis, whose utility for metabolic research mirrors that of RefSeq for genomic studies.

Like a reference genome, a reference metabolome would ideally incorporate continuous updates when more and better-quality data become available in association with the advancement of metabolomics technologies. As the field of plant metabolomics rapidly evolves, updates and expansion of RefMetaPlant will be crucial to keep pace with developments in the field. RefMetaPlant aims to become a centralized platform for plant metabolomic research. Its continuous expansion to include more organisms and metabolites is critically important to ensure its relevance, and to facilitate cross-referencing of information across studies.

RefMetaPlant database also serves as a collaborative platform, promoting knowledge/data sharing utilities and cross-field collaborative research. Besides housing reference metabolomes for >150 plants species that are available for online analysis and download, a data submission utility is provided for cross-team collaborations and for updating existing reference metabolomes or adding additional plant species. We seek to establish a secure and reliable system to facilitate big data exchange and to support the classification, collection, integration, quality control, archiving and 'one-stop' management, of plant metabolome data. Scientists from interdisciplinary fields can contribute their data, expanding the scope and depth of RefMetaPlant resources. Thus, RefMetaPlant will foster new collaborations in the field of metabolomic research, benefiting the research community through enhancing the distribution of collective knowledge.

In conclusion, the establishment of a plant reference metabolome database represents a significant milestone in plant metabolomics research. RefMetaPlant provides an invaluable resource, contributing to a comprehensive understanding of plant metabolism across diverse plant species, empowering researchers to delve deeper into the intricacies of plant metabolism.

Data availability

RefMetaPlant, help materials and related tools/resources are freely available at <https://www.biosino.org/RefMetaDB>, and are also accessible through <http://119.78.67.203/RefMetaDB>. This website is free for academic (non-commercial) applications, and is open to all users and no login or password is required. The metabolites data also have been deposited in

the BioMed Big Data Center (<https://www.biosino.org/node/>) under accession number OEP004638.

Supplementary data

Supplementary Data are available at NAR Online.

Acknowledgements

We thank Drs Yonghong Hu, Jianbing Yan, Chikuang Wen, Lin Xu for providing many of the plant samples, and Dr Jeremy Murray for language editing.

Funding

Strategic Priority Research Program of Chinese Academy of Sciences [XDA24010400]; National Natural Science Foundation of China [32270719, 31900470, 32200093]; Earmarked Fund for China Agriculture Research System [CARS-11]; Natural Science Foundation of Jiangsu Province [BK20220307]; Shanghai Post-doctoral Excellence Program [2021387]; Special funds for scientific research and development of Institute of Agricultural Sciences for Lixiahe Region in Jiangsu [SJ(22)107]. Funding for open access charge: Strategic Priority Research Program of Chinese Academy of Sciences [XDA24010400]; National Natural Science Foundation of China [32270719].

Conflict of interest statement

None declared.

References

- Zhao,Y.-Y. and Lin,R.-C. (2014) In: Makowski,G.S. (ed.) *Advances in Clinical Chemistry*. Elsevier, Vol. 65, pp. 69–89.
- Tauler,R. and Brown,S. (2009) *Comprehensive Chemometrics: Chemical and Biochemical Data Analysis*. Elsevier Science.
- Saito,K. and Matsuda,F. (2010) Metabolomics for functional genomics, systems biology, and biotechnology. *Annu. Rev. Plant Biol.*, **61**, 463–489.
- Dixon,R.A. and Strack,D. (2003) Phytochemistry meets genome analysis, and beyond. *Phytochemistry*, **62**, 815–816.
- de Souza,L.P., Alseekh,S., Scossa,F. and Fernie,A.R. (2021) Ultra-high-performance liquid chromatography high-resolution mass spectrometry variants for metabolomics research. *Nat. Methods*, **18**, 733–746.
- Alvarez-Rivera,G., Ballesteros-Vivas,D., Parada-Alfonso,F., Ibanez,E. and Cifuentes,A. (2019) Recent applications of high resolution mass spectrometry for the characterization of plant natural products. *Trac-Trends Anal. Chem.*, **112**, 87–101.
- Kusano,M., Yang,Z.G., Okazaki,Y., Nakabayashi,R., Fukushima,A. and Saito,K. (2015) Using metabolomic approaches to explore chemical diversity in rice. *Mol. Plant*, **8**, 58–67.
- Horai,H., Arita,M., Kanaya,S., Nihei,Y., Ikeda,T., Suwa,K., Ojima,Y., Tanaka,K., Tanaka,S., Aoshima,K., et al. (2010) MassBank: a public repository for sharing mass spectral data for life sciences. *J. Mass Spectrom.*, **45**, 703–714.
- Ara,T., Sakurai,N., Suzuki,H., Aoki,K., Saito,K. and Shibata,D. (2021) MassBase: a large-scaled depository of mass spectrometry datasets for metabolome analysis. *Plant Biotechnol.*, **38**, 167–171.
- Wishart,D.S., Guo,A., Oler,E., Wang,F., Anjum,A., Peters,H., Dizon,R., Sayeeda,Z., Tian,S., Lee,B.L., et al. (2021) HMDB 5.0: the Human Metabolome Database for 2022. *Nucleic Acids Res.*, **50**, D622–D631.
- Haug,K., Cochrane,K., Nainala,V.C., Williams,M., Chang,J.K., Jayaseelan,K.V. and O'Donovan,C. (2020) MetaboLights: a resource evolving in response to the needs of its scientific community. *Nucleic Acids Res.*, **48**, D440–D444.
- Sud,M., Fahy,E., Cotter,D., Azam,K., Vaidvelu,I., Burant,C., Edison,A., Fiehn,O., Higashi,R., Nair,K.S., et al. (2016) Metabolomics Workbench: an international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. *Nucleic Acids Res.*, **44**, D463–D470.
- Sakurai,N., Yamazaki,S., Suda,K., Hosoki,A., Akimoto,N., Takahashi,H., Shibata,D. and Aoki,Y. (2023) The Thing Metabolome Repository family (XMRs): comparable untargeted metabolome databases for analyzing sample-specific unknown metabolites. *Nucleic Acids Res.*, **51**, D660–D677.
- Fukushima,A., Takahashi,M., Nagasaki,H., Aono,Y., Kobayashi,M., Kusano,M., Saito,K., Kobayashi,N. and Arita,M. (2022) Development of RIKEN Plant Metabolome MetaDatabase. *Plant Cell Physiol.*, **63**, 433–440.
- Xue,J.C., Guijas,C., Benton,H.P., Warth,B. and Siuzdak,G. (2020) METLIN MS2 molecular standards database: a broad chemical and biological resource. *Nat. Methods*, **17**, 953–954.
- Fiehn,O., Wohlgemuth,G. and Scholz,M. (2005) *Data Integration in the Life Sciences: Second International Workshop, DILS 2005, San Diego, CA, USA, July 20-22, 2005. Proceedings 2*. Springer, pp. 224–239.
- O'Leary,N.A., Wright,M.W., Brister,J.R., Ciufu,S., McVeigh,D.H.R., Rajput,B., Robbertse,B., Smith-White,B., Ako-Adjei,D., Astashyn,A., et al. (2016) Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Res.*, **44**, D733–D745.
- Iijima,Y., Nakamura,Y., Ogata,Y., Tanaka,K., Sakurai,N., Suda,K., Suzuki,T., Suzuki,H., Okazaki,K., Kitayama,M., et al. (2008) Metabolite annotations based on the integration of mass spectral information. *Plant J.*, **54**, 949–962.
- Moco,S., Bino,R.J., Vorst,O., Verhoeven,H.A., de Groot,J., van Beek,T.A., Vervoort,J. and de Vos,C.H.R. (2006) A liquid chromatography-mass spectrometry-based metabolome database for tomato. *Plant Physiol.*, **141**, 1205–1218.
- Grafahrend-Belau,E., Weise,S., Koschützki,D., Scholz,U., Junker,B.H. and Schreiber,F. (2008) MetaCrop: a detailed database of crop plant metabolism. *Nucl. Acids Res.*, **36**, D954–D958.
- Deborde,C. and Jacob,D. (2014) MeRy-B, a metabolomic database and knowledge base for exploring plant primary metabolism. *Methods Mol. Biol.*, **1083**, 3–16.
- Singla,R.K., De,R., Efferth,T., Mezzetti,B., Uddin,S., M., S., Ntie-Kang,F., Wang,D., Schultz,F., Kharat,K.R., et al. (2023) The International Natural Product Sciences Taskforce (INPST) and the power of Twitter networking exemplified through #INPST hashtag analysis. *Phytomedicine*, **108**, 154520.
- Sayers,E.W., Bolton,E.E., Brister,J.R., Canese,K., Chan,J., Comeau,D.C., Connor,R., Funk,K., Kelly,C., Kim,S., et al. (2022) Database resources of the national center for biotechnology information. *Nucleic Acids Res.*, **50**, D20–D26.
- Li,X.T., Zhou,H.X., Xiao,N., Wu,X.T., Shan,Y.H., Chen,L.X., Wang,C.T., Wang,Z.X., Huang,J.R., Li,A.H., et al. (2022) Expanding the Coverage of Metabolic Landscape in Cultivated Rice with Integrated Computational Approaches. *Genomics Proteomics Bioinformatics*, **20**, 702–714.
- Duan,L.X., Molnar,I., Snyder,J.H., Shen,G.A. and Qi,X.Q. (2016) Discrimination and quantification of true biological signals in metabolomics analysis based on liquid chromatography-mass spectrometry. *Mol. Plant*, **9**, 1217–1220.
- Feunang,Y.D., Eisner,R., Knox,C., Chepelev,L., Hastings,J., Owen,G., Fahy,E., Steinbeck,C., Subramanian,S., Bolton,E., et al. (2016) ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. *J. Cheminformatics*, **8**, 61.
- Sawada,Y., Nakabayashi,R., Yamada,Y., Suzuki,M., Sato,M., Sakata,A., Akiyama,K., Sakurai,T., Matsuda,F., Aoki,T., et al.

- (2012) RIKEN tandem mass spectral database (ReSpect) for phytochemicals: a plant-specific MS/MS-based data resource and database. *Phytochemistry*, **82**, 38–45.
28. Kind, T., Wohlgemuth, G., Lee, D.Y., Lu, Y., Palazoglu, M., Shahbaz, S. and Fiehn, O. (2009) FiehnLib: mass spectral and retention index libraries for metabolomics based on quadrupole and time-of-flight gas chromatography/mass spectrometry. *Anal. Chem.*, **81**, 10038–10048.
 29. Allen, F., Pon, A., Wilson, M., Greiner, R. and Wishart, D. (2014) CFM-ID: a web server for annotation, spectrum prediction and metabolite identification from tandem mass spectra. *Nucl. Acids Res.*, **42**, W94–W99.
 30. Kanehisa, M. and Goto, S. (2000) KEGG: kyoto Encyclopedia of Genes and Genomes. *Nucl. Acids Res.*, **28**, 27–30.
 31. Afendi, F.M., Okada, T., Yamazaki, M., Hirai-Morita, A., Nakamura, Y., Nakamura, K., Ikeda, S., Takahashi, H., Altaf-Ul-Amin, M., Darusman, L.K., *et al.* (2012) KnapSack family databases: integrated metabolite-plant species databases for multifaceted plant research. *Plant Cell Physiol.*, **53**, e1.
 32. Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.* (2023) PubChem 2023 update. *Nucl. Acids Res.*, **51**, D1373–D1380.
 33. Gu, J., Gui, Y., Chen, L., Yuan, G., Lu, H.-Z. and Xu, X. (2013) Use of natural products as chemical library for drug discovery and network pharmacology. *PLoS One*, **8**, e62839.
 34. O'Boyle, N.M., Banck, M., James, C.A., Morley, C., Vandermeersch, T. and Hutchison, G.R. (2011) Open Babel: an open chemical toolbox. *J. Cheminformatics*, **3**, 33.
 35. Geer, L.Y., Marchler-Bauer, A., Geer, R.C., Han, L., He, J., He, S., Liu, C., Shi, W. and Bryant, S.H. (2010) The NCBI BioSystems database. *Nucl. Acids Res.*, **38**, D492–D496.
 36. Degtyarenko, K., de Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcantara, R., Darsow, M., Guedj, M. and Ashburner, M. (2008) ChEBI: a database and ontology for chemical entities of biological interest. *Nucl. Acids Res.*, **36**, D344–D350.
 37. Sud, M., Fahy, E., Cotter, D., Brown, A., Dennis, E.A., Glass, C.K., Merrill, A.H. Jr., Murphy, R.C., Raetz, C.R., Russell, D.W., *et al.* (2007) LMSD: LIPID MAPS structure database. *Nucleic Acids Res.*, **35**, D527–D532.
 38. Bienfait, B. and Ertl, P. (2013) JSME: a free molecule editor in JavaScript. *J. Cheminformatics*, **5**, 24.
 39. Willighagen, E.L., Mayfield, J.W., Alvarsson, J., Berg, A., Carlsson, L., Jeliazkova, N., Kuhn, S., Pluskal, T., Rojas-Cherto, M., Spjuth, O., *et al.* (2017) The Chemistry Development Kit (CDK) v2.0: atom typing, depiction, molecular formulas, and substructure searching. *J. Cheminformatics*, **9**, 33.
 40. Sokolow, S., Karnofsky, J. and Gustafson, P. (1978) The Finnigan library search program: Finnigan application report 2. *Finnigan Corp.*, **2**, 1–45.
 41. Smith, C.A., Want, E.J., O'Maille, G., Abagyan, R. and Siuzdak, G. (2006) XCMS: processing mass spectrometry data for metabolite profiling using nonlinear peak alignment, matching, and identification. *Anal. Chem.*, **78**, 779–787.
 42. Bodenhofer, U., Kothmeier, A. and Hochreiter, S. (2011) APCluster: an R package for affinity propagation clustering. *Bioinformatics*, **27**, 2463–2464.
 43. Federhen, S. (2015) Type material in the NCBI Taxonomy Database. *Nucl. Acids Res.*, **43**, D1086–D1098.
 44. Letunic, I. and Bork, P. (2021) Interactive Tree Of Life (iTOL) v5: an online tool for phylogenetic tree display and annotation. *Nucleic Acids Res.*, **49**, W293–W296.
 45. Ma, B., Zhang, L., Gao, Q.F., Wang, J.M., Li, X.Y., Wang, H., Liu, Y., Lin, H., Liu, J.Y., Wang, X., *et al.* (2021) A plasma membrane transporter coordinates phosphate reallocation and grain filling in cereals. *Nat. Genet.*, **53**, 906–915.
 46. Zhang, H., Chen, T.T., Wang, Z.Q., Yang, J.C. and Zhang, J.H. (2010) Involvement of cytokinins in the grain filling of rice under alternate wetting and drying irrigation. *J. Exp. Bot.*, **61**, 3719–3733.
 47. Min, X.M., Xu, H.L., Huang, F.L., Wei, Y.D., Lin, W.X. and Zhang, Z.X. (2021) GC-MS-based metabolite profiling of key differential metabolites between superior and inferior spikelets of rice during the grain filling stage. *BMC Plant Biol.*, **21**, 439.
 48. Xu, S.B., Li, T., Deng, Z.Y., Chong, K., Xue, Y.B. and Wang, T. (2008) Dynamic proteomic analysis reveals a switch between central carbon metabolism and alcoholic fermentation in rice filling grains. *Plant Physiol.*, **148**, 908–925.
 49. Chambers, M.C., Maclean, B., Burke, R., Amodei, D., Ruderman, D.L., Neumann, S., Gatto, L., Fischer, B., Pratt, B., Egertson, J., *et al.* (2012) A cross-platform toolkit for mass spectrometry and proteomics. *Nat. Biotechnol.*, **30**, 918–920.
 50. Riedelsheimer, C., Lisec, J., Czedik-Eysenberg, A., Sulpice, R., Flis, A., Grieder, C., Altmann, T., Stitt, M., Willmitzer, L. and Melchinger, A.E. (2012) Genome-wide association mapping of leaf metabolic profiles for dissecting complex traits in maize. *Proc. Nat. Acad. Sci. U.S.A.*, **109**, 8872–8877.
 51. Wen, W.W., Li, D., Li, X., Gao, Y.Q., Li, W.Q., Li, H.H., Liu, J., Liu, H.J., Chen, W., Luo, J., *et al.* (2014) Metabolome-based genome-wide association study of maize kernel leads to novel biochemical insights. *Nat. Commun.*, **5**, 3438.
 52. Deng, M., Zhang, X.H., Luo, J.Y., Liu, H.J., Wen, W.W., Luo, H.B., Yan, J.B. and Xiao, Y.J. (2020) Metabolomics analysis reveals differences in evolution between maize and rice. *Plant J.*, **103**, 1710–1722.
 53. Li, Z., Zhu, A.D., Song, Q.X., Chen, H.Y., Harmon, F.G. and Chen, Z.J. (2020) Temporal regulation of the metabolome and proteome in photosynthetic and photorespiratory pathways contributes to maize heterosis. *Plant Cell*, **32**, 3706–3722.
 54. Liang, X., Liu, S., Wang, T., Li, F., Cheng, J., Lai, J., Qin, F., Li, Z., Wang, X. and Jiang, C. (2021) Metabolomics-driven gene mining and genetic improvement of tolerance to salt-induced osmotic stress in maize. *New Phytol.*, **230**, 2355–2370.
 55. Forrister, D.L., Endara, M.J., Soule, A.J., Younkin, G.C., Mills, A.G., Lokvam, J., Dexter, K.G., Pennington, R.T., Kidner, C.A., Nicholls, J.A., *et al.* (2023) Diversity and divergence: evolution of secondary metabolism in the tropical tree genus *Inga*. *New Phytol.*, **237**, 631–642.
 56. Wu, Q., Tong, W., Zhao, H.J., Ge, R.H., Li, R.P., Huang, J., Li, F.D., Wang, Y.L., Mallano, A.L., Deng, W.W., *et al.* (2022) Comparative transcriptomic analysis unveils the deep phylogeny and secondary metabolite evolution of 116 Camellia plants. *Plant J.*, **111**, 406–421.
 57. Zhang, Y., Deng, T., Sun, L., Landis, J.B., Moore, M.J., Wang, H., Wang, Y., Hao, X., Chen, J., Li, S., *et al.* (2021) Phylogenetic patterns suggest frequent multiple origins of secondary metabolites across the seed-plant 'tree of life'. *Natl. Sci. Rev.*, **8**, nwa105.
 58. Zhang, R., Wang, Y.H., Jin, J.J., Stull, G.W., Bruneau, A., Cardoso, D., De Queiroz, L.P., Moore, M.J., Zhang, S.D., Chen, S.Y., *et al.* (2020) Exploration of plastid phylogenomic conflict yields new insights into the deep relationships of leguminosae. *Syst. Biol.*, **69**, 613–622.
 59. Liu, W., Feng, Y., Yu, S., Fan, Z., Li, X., Li, J. and Yin, H. (2021) The flavonoid biosynthesis network in plants. *Int. J. Mol. Sci.*, **22**, 12824.
 60. Shen, N., Wang, T.F., Gan, Q., Liu, S., Wang, L. and Jin, B. (2022) Plant flavonoids: classification, distribution, biosynthesis, and antioxidant activity. *Food Chem.*, **383**, 132531.
 61. Nabavi, S.M., Samec, D., Tomczyk, M., Milella, L., Russo, D., Habtemariam, S., Sutar, I., Rastrelli, L., Daglia, M., Xiao, J.B., *et al.* (2020) Flavonoid biosynthetic pathways in plants: versatile targets for metabolic engineering. *Biotechnol. Adv.*, **38**, 107316.
 62. Veitch, N.C. (2007) Isoflavonoids of the leguminosae. *Nat. Prod. Rep.*, **24**, 417–464.
 63. Bulut, M., Wendenburg, R., Bitocchi, E., Bellucci, E., Kroc, M., Gioia, T., Susek, K., Papa, R., Fernie, A.R. and Alseekh, S. (2023) A comprehensive metabolomics and lipidomics atlas for the legumes common bean, chickpea, lentil and lupin. *Plant J.*, **2023**, <https://doi.org/10.1111/tpj.16329>.

64. Tohge,T., de Souza,L.P. and Fernie,A.R. (2018) On the natural diversity of phenylacylated-flavonoid and their in planta function under conditions of stress. *Phytochem. Rev.*, **17**, 279–290.
65. Butkute,B., Padaraukas,A., Ceseviciene,J., Taujenis,L. and Norkevicien,E. (2018) Phytochemical composition of temperate perennial legumes. *Crop Pasture Sci.*, **69**, 1020–1030.
66. Jones,P. and Vogt,T. (2001) Glycosyltransferases in secondary plant metabolism: tranquilizers and stimulant controllers. *Planta*, **213**, 164–174.
67. Bowles,D., Lim,E.K., Poppenberger,B. and Vaistij,F.E. (2006) Glycosyltransferases of lipophilic small molecules. *Annu. Rev. Plant Biol.*, **57**, 567–597.